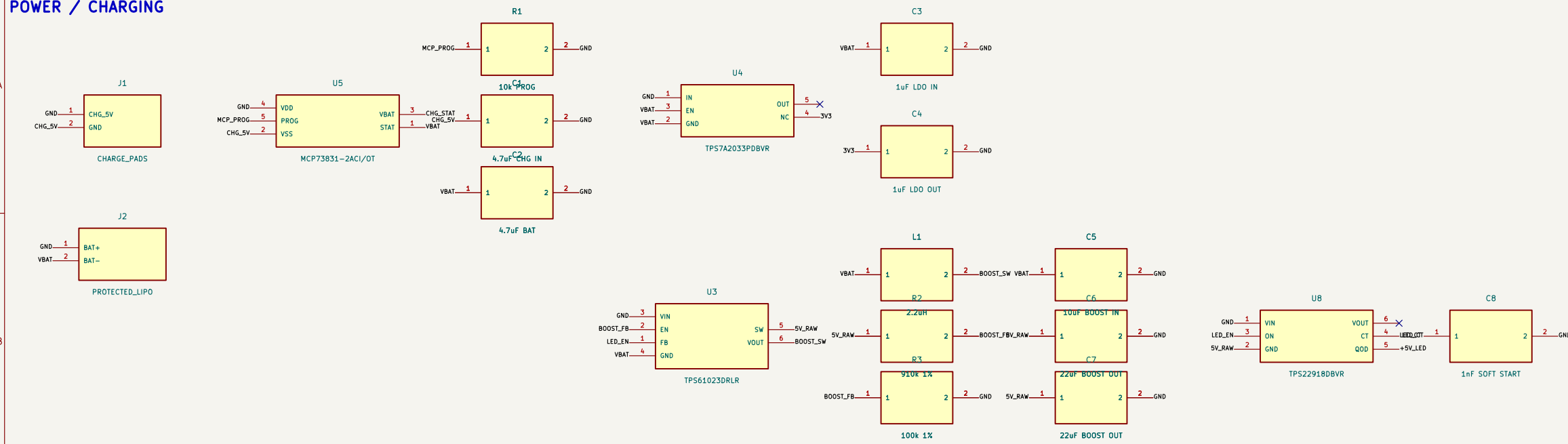
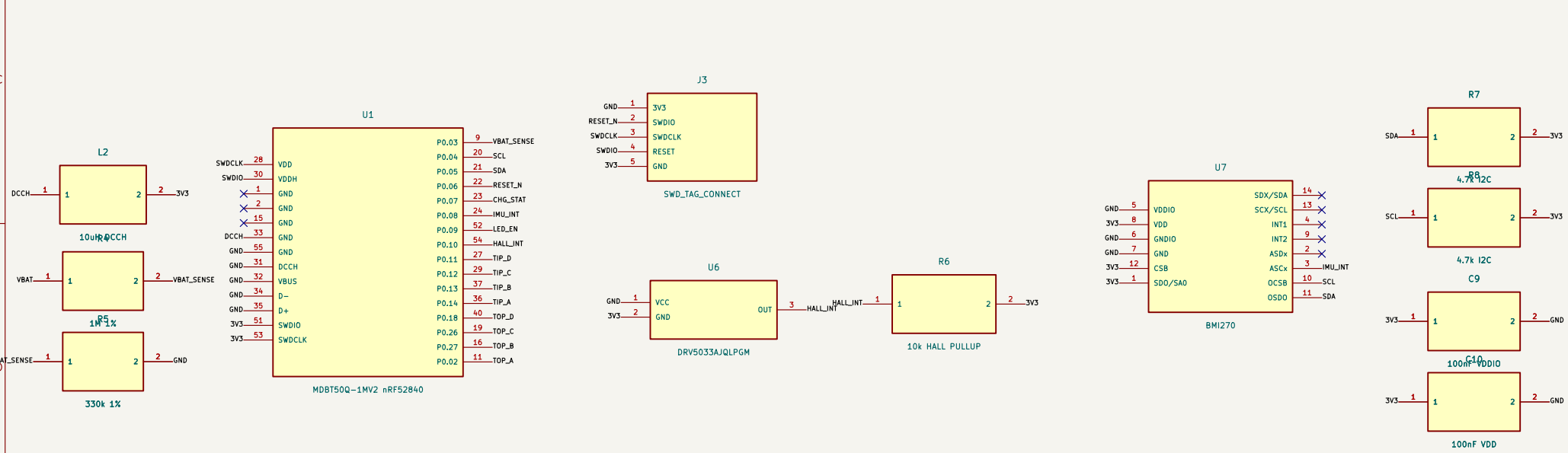


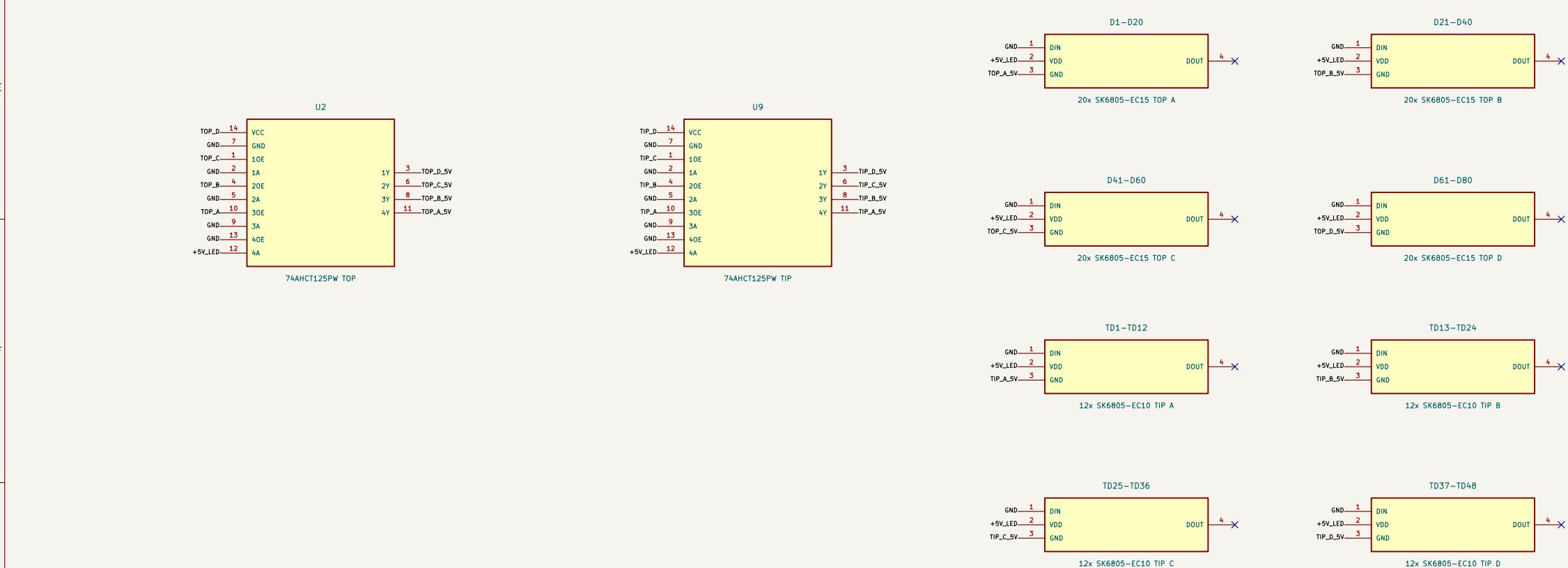
POWER / CHARGING



CONTROL / SENSORS



LEVEL SHIFT / LED DATA



ENGINEERING REVIEW NOTES

1. This is the first editable print schematic; the earlier v0.04 PDF was an architecture document and was incorrectly named.
2. Repeated LED strings are shown as chain blocks. Expand/anoate all 128 LED references when synchronizing schematic and PCB.
3. Protected LS150 required. No battery is to be connected before charge polarity and protection are verified.
TSP56102201 is a 12V 1000mA 1.5W LED driver. It is not a 12V 1000mA 1.5W LED driver. It is a 12V 1000mA 1.5W LED driver.
5. TSP52291B is limited to 2 A continuous. Firmware must cap brightness/current; thermal and transient tests are mandatory.
6. Verify every footprint against the purchased MPN, especially BM1702 LG and SK6805 LED variants.
7. Be aware of the 7.6V antenna cutoff frequency and dynamic impedance, and dynamic impedance, and dynamic impedance.
8. This review file does not authorize fabrication. Release only after independent schematic review, ERC/DRC cleanup and DFM.

All named labels are electrical nets. Repeated LED strings are equivalent chain blocks.
4 arms x 20 top LEDs + 4 replaceable tip boards x 12 LEDs.
NOT FOR FABRICATION. Validate footprints, current limits, thermal design and antenna keep-out.

Sheet: /		
File: smartspinner_main_v0.5.kicad_sch		
Title: SMARTSPINNER v0.5 – PRINCIPLE SCHEMATIC FOR ENGINEERING REVIEW		
Size: A2	Date: 2026–09–22	Rev: v0.5–review
KiCad E.D.A. 10.0.6		Id: 1/1